Step 1: Define Activities & Dependencies

Code	Activity	Predecessor(s)
A	Requirement Analysis	—
B	System & UI Design	A
C	Database Design	B
D	Patient Module Development	B
E	Doctor & Staff Module Development	C
F	Billing & Room Module Integration	D, E
G	System Testing	F

Step 2: Time Estimation

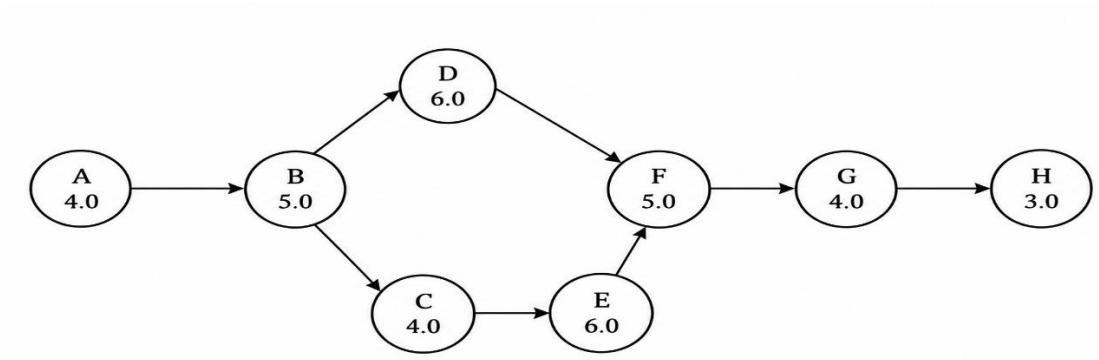
Using the PERT formula:

TE= O+4M+P/6

Where:
O = Optimistic time
M = Most Likely time
P = Pessimistic time

Activity	O (days)	M (days)	P (days)	TE (days)
A	2	4	6	4.0
B	3	5	7	5.0
C	2	4	6	4.0
D	3	6	9	6.0
E	4	6	8	6.0
F	2	5	8	5.0
G	3	4	5	4.0
H	1	3	5	3.0

Step 3: Draw PERT/CPM Network Diagram



Critical Path

A → B → C → E → F → G → H

Project Completion Time = 31 Days

Step 4: Gantt Chart / Timeline Representation

Activity	Duration	Start	End
A	4 days	Day 1	Day 4
B	5 days	Day 5	Day 9
C	4 days	Day 10	Day 13
D	6 days	Day 10	Day 15
E	6 days	Day 14	Day 19
F	5 days	Day 20	Day 24
G	4 days	Day 25	Day 28
H	3 days	Day 29	Day 31

